CSE475 / TASK 1

Bonn EEG exploration and related work

Track 1 - Graph Neural Networks | 15 October 2026

Submitted to

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Group 02

1. Dataset and objective

This study audits 500 EEG segments supplied in five archives from Neurophysics Bonn [D1]. It explores numerical features and develops a controlled graph-learning question. No classifier is trained in Task 1. All empirical statistics below were calculated from the uploaded files; published model results are labelled separately.

Dataset / area	Bonn EEG (Andrzejak et al., 2001); EEG signal classification
Shape / axes	500 single-channel segments x 4,097 time-ordered amplitude values; 2,048,500 values total. A segment is not a patient.
Sampling / duration	173.61 Hz; N/fs = 23.60 s per segment. Acquisition bandwidth stated as 0.5-85 Hz.
Feature table	19 engineered numerical descriptors per segment
Subjects / sources	No per-file patient ID or parent-recording mapping supplied. Subject count and independent acquisition count cannot be recovered from filenames.
Data discrepancy	Page states 4,096 samples; all supplied signals have 4,097. Retain the actual values.
Analysis scope	Audit all 500; detailed feature EDA uses 300 development segments. Reserve 100 validation + 100 test segments from feature exploration.

Class / set	Meaning	Total / EDA
Z / A	Healthy; eyes open; scalp EEG	100 / 60
O / B	Healthy; eyes closed; scalp EEG	100 / 60
N / C	Seizure-free; contralateral hippocampal formation	100 / 60
F / D	Seizure-free; epileptogenic zone	100 / 60
S / E	Seizure activity; intracranial EEG	100 / 60

Each five-class category is 20% of the collection. A binary S-versus-rest task would instead have a 1:4 imbalance. Healthy and patient classes differ in acquisition context, which can contribute to separability. The task is segment classification, not future-seizure prediction.

Scope limitation: the fixed 60/20/20 allocation is segment-level only. It does not establish patient or parent-recording independence. Resolve this with the instructor before treating it as the required subject-wise modelling protocol.

2. Statistics and data quality

A-C | All-file audit; feature statistics on the development subset

Audit result	Measured value
Valid signal files	500; 100 per class
Malformed / missing / infinite	0 malformed files; 0 nonfinite values
Constant / exact duplicate	0 constant segments; 0 exact duplicate numerical signals
Engineered missing / near-constant	0 missing values; 0 features with >=99% identical values
Outlier flags	557 feature-segment flags across 132 of 300 development segments (pooled 1.5-IQR rule)

Selected feature statistics (300 segments)

Feature	Mean	Median	Std	Min	Max
rms	108.910	64.429	116.934	19.755	590.987
std	103.002	54.187	118.761	18.609	591.034
line length	23.919	13.284	30.150	3.246	209.146
entropy	0.609	0.603	0.081	0.433	0.809
alpha mass	0.214	0.164	0.171	0.023	0.783

Complete quartiles, skewness, excess kurtosis, class-wise summaries and outlier flags are included in the notebook and exported CSV tables. Here std in the first column is the within-segment feature; Std in the header is its across-segment variability.

Preprocessing decisions

Keep all segments: outliers are not synonymous with corruption. A high-amplitude seizure segment can legitimately be unusual relative to the pooled population. Exact-duplicate absence does not exclude near-duplicates or repeated observations from the same patient. No interpolation, resampling, extra filtering or per-record amplitude normalization was applied.

The 19 features comprise 13 time-domain summaries and six spectral summaries. Welch PSD uses fs=173.61 Hz and nperseg=256. Relative delta/theta/alpha/beta/gamma masses use [0.5,4), [4,8), [8,13), [13,30), [30,45) Hz and divide discrete band PSD sums by the total PSD sum. Entropy is normalized by log2(number of PSD bins).

The download page states that the manuscript's 40-Hz low-pass filter was not applied to the downloadable time series. Our raw EDA therefore does not assume that filter. Later model preprocessing must be fixed in advance and fitted only on the training partition. Scaling and PCA here use only development data.

3. Distributions and balance

D-E | Each development class contains 60 segments; no resampling

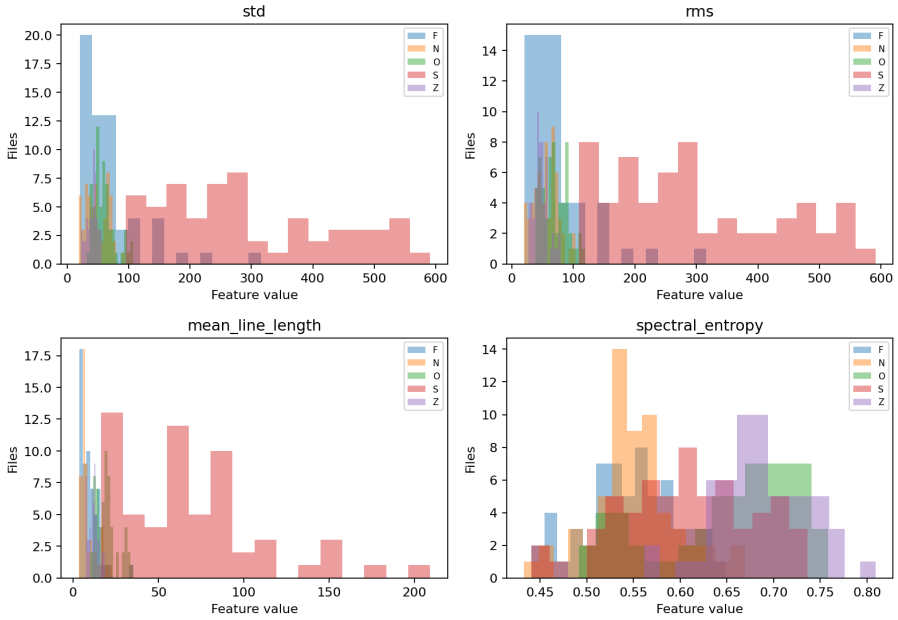


Figure 1. Development feature histograms. S extends towards larger amplitude and line-length values, whereas F/N and O/Z share substantial ranges. The shape differences motivate several complementary features rather than a single threshold.

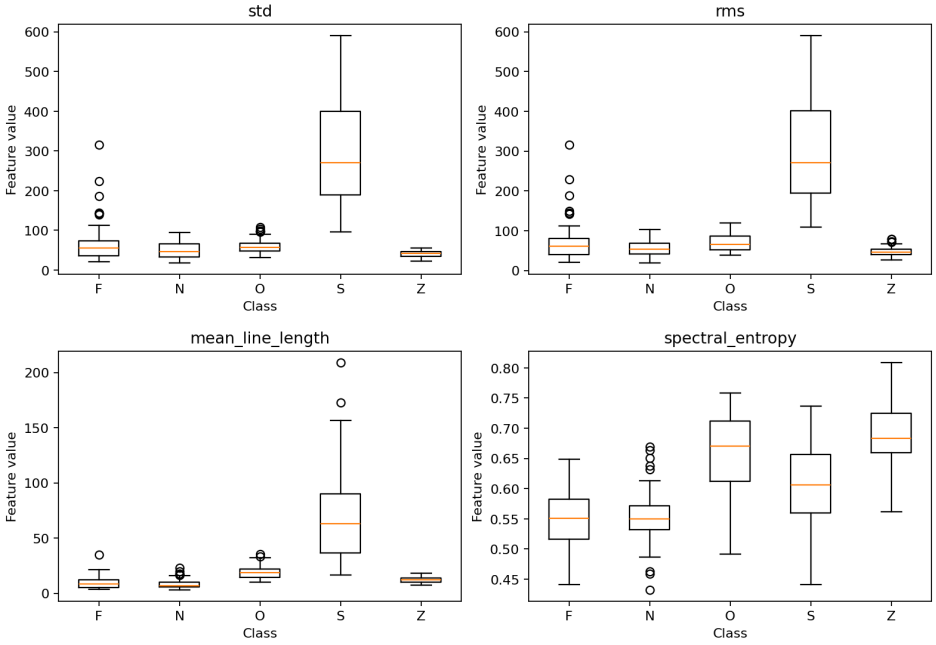


Figure 2. S median RMS is 271.43 versus 46.00-66.00 for the other classes. F has high-amplitude observations overlapping S. F/N spectral-entropy medians are 0.5506/0.5498; O/Z are 0.6703/0.6836. This overlap makes five-class classification more demanding than a simple healthy-versus-seizure comparison.

Violin plots and one raw waveform per class are included in the executed notebook. They show broad within-class variation; representative traces are not population estimates. O has median relative alpha mass 0.431 versus Z 0.241; this is an association, not a causal conclusion.

4. Correlation and projection

F-H | Development-only feature relationships and interactive views

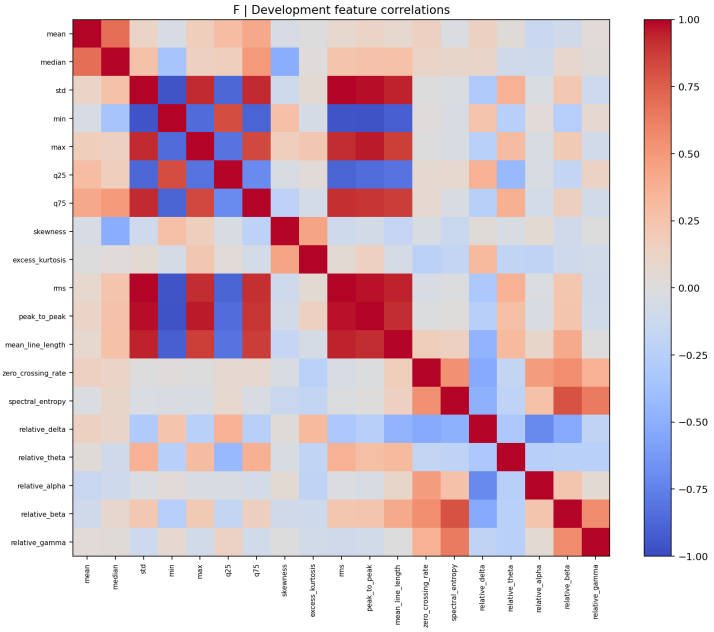


Figure 3. Seven feature pairs satisfy |r| >= 0.95. RMS/std correlation is 0.99594, and RMS/peak-to-peak is 0.97461. These amplitude summaries are partly redundant. A later feature ablation should compare retaining versus removing redundant features within each training fold.

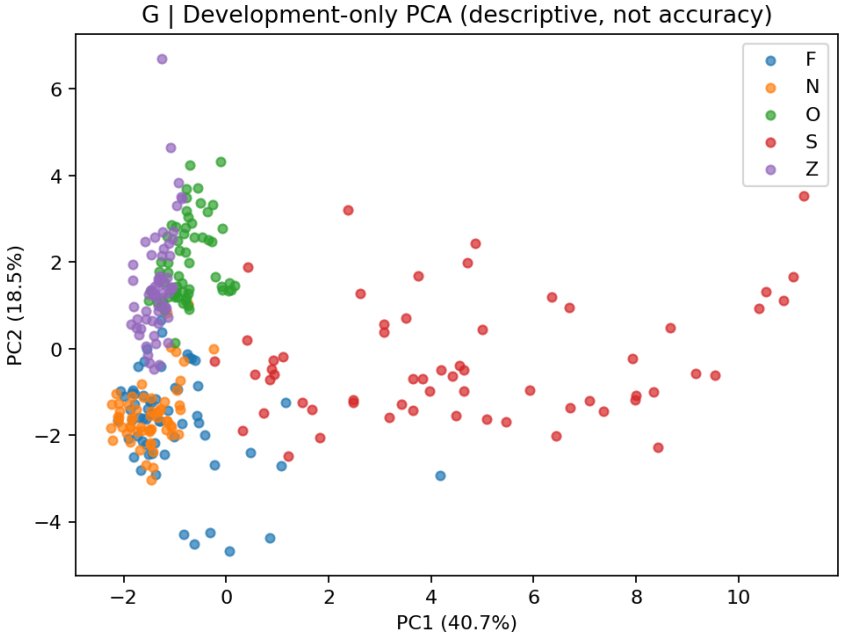


Figure 4. PC1 explains 40.71% and PC2 18.48% (59.19% combined); PC3 adds 11.80%. S generally extends towards higher PC1, while F/N and O/Z overlap. Standardization was fitted on the 300 development segments. Projection separation is not test accuracy.

Interactive PCA (2D/3D) and RMS HTML plots are included under outputs/task1/. Hover identifies each segment; zoom and legend controls help inspect overlap and extremes. They are self-contained and work offline. Correlation is between engineered features, not physical EEG channels.

5. Related work / 1 of 3

Five recent EEG/GNN papers; all numbers below are author-reported, not reproduced project results.

R1 | Self-Supervised Learning for Anomalous Channel Detection in EEG Graphs: Application to Seizure Analysis

Thi Kieu Khanh Ho; Narges Armanfard | 2023 (AAAI; preprint 2022)

Dataset / area	TUSZ v1.5.2, 19 channels; Seizure clip and channel detection
Method / GNN	EEG-CGS: GCN autoencoder, contrastive and reconstruction losses
Graph	Electrodes are nodes; distance, random, correlation and directed-transfer graph variants.
Evaluation	Normal-only training: 35,019 clips; held-out test: 8,848 clips including seizures.
Reported metrics	Combined graphs: clip F1 0.534, sensitivity 0.501, specificity 0.974 (Table 3).
Strength	Learns from normal data and compares graph variants.
Limitation	Channel-localization task requires multichannel signals; cannot transfer unchanged to a single channel.
Gap / relation	Contribution of temporal edges in single-channel graphs is not answered by this multichannel experiment. Motivates controlled edge-rule ablations; numbers are not Bonn targets.
Evidence	Full text, graph construction and Tables 1/3.

R2 | Dynamic GNNs for Precise Seizure Detection and Classification from EEG Data

Arash Hajisafi; Haowen Lin; Yao-Yi Chiang; Cyrus Shahabi | 2024 (arXiv version reviewed)

Dataset / area	TUSZ v1.5.2; Seizure detection and type classification
Method / GNN	NeuroGNN: modified residual GCN, BiGRU and hierarchical pooling
Graph	Electrodes and brain-region meta-nodes; temporal, spatial and semantic edges.
Evaluation	Separate test set; training portion divided 9:1 for training/validation; three random runs.
Reported metrics	With pretraining: detection AUROC 0.876 +/- 0.002; classification weighted F1 0.792 +/- 0.004 (Table 2).
Strength	Tests each graph component through ablation.
Limitation	Requires electrode geometry and brain-region information absent from isolated single-channel segments.
Gap / relation	Does temporal order alone add value when spatial information is unavailable? Supports matching feature inputs and changing one edge mechanism at a time.
Evidence	Full text, Sections 3-4 and Table 2.

5. Related work / 2 of 3

R3 | Sequential graph convolutional network and DeepRNN based hybrid framework for epileptic seizure detection from EEG signal

Ferdaus Anam Jibon; A. R. Jamil Chowdhury; Mahadi Hasan Miraz; Hwang Ha Jin; Mayeen Uddin Khandaker; Sajia Sultana; Sifat Nur; Fazlul Hasan Siddiqui; AHM Kamal; Mohammad Salman; Ahmed A. F. Youssef | 2024

Dataset / area	CHB-MIT and TUH; Seizure detection
Method / GNN	Sequential GCN plus DeepRNN
Graph	Pearson-correlation graph; periodogram-based signal processing.
Evaluation	CHB-MIT patient-by-patient experiments: 5-fold CV, repeated 10 times. Subject-disjoint testing not established by this description.
Reported metrics	Table 2 accuracy: CHB-MIT 99.007%; TUH 98.08%.
Strength	Compares the hybrid against its individual components.
Limitation	Reporting inconsistencies: TUH sensitivity/specificity entries differ between Tables 2 and 5.
Gap / relation	Need auditable splits and consistent metric reporting before comparing architectures. Use reproducible split manifests and distinguish patient-specific from unseen-patient claims.
Evidence	Full text, Algorithm 1 and Tables 2/3/5. Results transcribed, not independently reproduced.

R4 | Multi-branch fusion graph neural network based on multi-head attention for childhood seizure detection

Yang Li; Yang Yang; Shangling Song; Hongjun Wang; Mengzhou Sun; Xiaoyun Liang; Penghui Zhao; Baiyang Wang; Na Wang; Qiyue Sun; Zijuan Han | 2024

Dataset / area	CHB-MIT; Childhood seizure detection
Method / GNN	MGCNA: multi-branch GCN with multi-head attention
Graph	Spatial-distance, functional-correlation and trainable graphs over channels.
Evaluation	Patient-specific 5-fold CV; patient-independent leave-one-subject-out evaluation.
Reported metrics	Table 1: patient-specific accuracy 99.32%, F1 97.76%. Patient-independent accuracy conflicts: 89.45% in Table 2 versus 85.45% in Table 4.
Strength	Separately studies patient-specific and patient-independent performance.
Limitation	Requires channel locations; reported cross-patient accuracy is internally inconsistent.
Gap / relation	Need simple graph ablations with explicit generalization limits. Supports reporting source grouping and computational cost rather than chasing a published accuracy.
Evidence	Full text, Sections 2.8/4.3 and Tables 1/2/4.

5. Related work / 3 of 3

R5 | Lightweight Graph Neural Network Framework for Seizure Detection Using Single-Channel EEG

Shobha Sharma; Tapan Kumar Gandhi | 2026 (online 17 April; PReMI 2025 proceedings)

Dataset / area	Bonn, five categories Z/N/F/S/O; EEG segment classification
Method / GNN	Lightweight GCN
Graph	Eight non-overlapping temporal segments per recording; feature-similarity edges.
Evaluation	Ten random seeds; split proportions and subject grouping not disclosed in the accessible abstract.
Reported metrics	Across seeds: accuracy 0.87 +/- 0.0147; macro-F1 0.8681 +/- 0.0157. Best-seed accuracy 91%.
Strength	Direct precedent for a single-channel temporal-segment graph.
Limitation	Only publisher abstract accessible; exact features, edges and split cannot be verified.
Gap / relation	The accessible evidence does not isolate the value of temporal-adjacency edges versus similarity edges. Closest prior work. Propose a controlled extension, not a claim that Bonn GNNs are new.
Evidence	Publisher abstract only; full chapter is subscription content.

6. Research gap and testable question

The reviewed multichannel GNNs use electrode relationships that cannot be copied directly to these single-channel segments. R5 already constructs a Bonn graph from eight temporal parts, so applying a GNN to Bonn is not a new contribution. The accessible evidence does not isolate temporal adjacency from feature-similarity edges; this motivates a controlled project experiment, not a claim of proven global novelty.

Question	On a fixed, documented segment-level Bonn partition, does adding temporal-neighbour edges to similarity edges improve macro-F1 over temporal-only and similarity-only GCNs under identical features and training budget?
Proposed graph	One released EEG segment per graph; eight contiguous parts as nodes. numpy.array_split preserves 4,097 values: 513 + seven parts of 512.
Node / edge design	19 descriptors per node; temporal chain, feature-similarity k-nearest neighbours, or their union. Scale features using training nodes only; select k on validation data.
Controlled comparison	Same splits, node features, GCN size, optimization budget and seeds. Also compare Logistic Regression, SVM and Random Forest on segment features.
Outcome	Macro-F1, class recall, accuracy, time and parameter count; a negative result is valid.

Temporal edges encode order, not anatomical brain connectivity. Keep every edge within its original segment and keep all derived parts together. Subject-level generalization remains unverified because patient/parent-recording metadata are absent. The instructor must decide whether this limitation and temporal graph satisfy Track 1 before later modelling.

7. Conclusions and references

Task 1 findings

The uploaded release contains 500 balanced single-channel segments and no detected missing/nonfinite values, constant signals or exact duplicates. Seizure segments show higher amplitude summaries, but the five classes overlap. Several amplitude features are highly correlated, and pooled outlier rules flag genuine class variation. These findings support careful feature scaling, controlled redundancy ablations and reporting class-wise errors in subsequent work.

The empirical EDA, executed notebook, five-paper table and research question are provided. No classification accuracy is claimed. Source/patient independence cannot be established from the supplied filenames. Team details appear on the cover. Group 02.

Reproducibility

Extract the package, install requirements.txt, and run code/task1/Group02_Bonn_task1_eda.ipynb from top to bottom. The package includes the original numerical signals, archive hashes, file manifest, fixed segment allocation, feature tables, charts and offline interactive views. Seeds 475 and 476 define the provisional allocation. The README maps outputs to rubric A-H.

References

[R1] Thi Kieu Khanh Ho et al. (2023 (AAAI; preprint 2022)). Self-Supervised Learning for Anomalous Channel Detection in EEG Graphs: Application to Seizure Analysis.
https://arxiv.org/html/2208.07448v4

[R2] Arash Hajisafi et al. (2024 (arXiv version reviewed)). Dynamic GNNs for Precise Seizure Detection and Classification from EEG Data.
https://arxiv.org/html/2405.09568v1

[R3] Ferdaus Anam Jibon et al. (2024). Sequential graph convolutional network and DeepRNN based hybrid framework for epileptic seizure detection from EEG signal.
https://journals.sagepub.com/doi/10.1177/20552076241249874

[R4] Yang Li et al. (2024). Multi-branch fusion graph neural network based on multi-head attention for childhood seizure detection.
https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2024.1439607/full

[R5] Shobha Sharma et al. (2026 (online 17 April; PReMI 2025 proceedings)). Lightweight Graph Neural Network Framework for Seizure Detection Using Single-Channel EEG.
https://link.springer.com/chapter/10.1007/978-3-032-18477-1_47

[D1] Neurophysics Bonn. EEG download page. https://neurophysicsbonn.de/downloads . Accessed 15 October 2026. User supplied z.zip, o.zip, n.zip, f.zip and s.zip from this source. No alternate dataset used.

[D2] Andrzejak RG, Lehnertz K, Mormann F, Rieke C, David P, Elger CE (2001). Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state. Physical Review E 64, 061907. https://doi.org/10.1103/PhysRevE.64.061907 . Dataset citation, additional to the five recent papers.

[C1] Supplied CSE475 project guide, Track 1 brief, Rules section 6.3, and Task 1 Classroom screenshot. Deliverable: data EDA plus related-work table and gap.